ED-FEED: An AI-Driven Intelligent Student Feedback Analysis and Insight Generation System

BASELINE MODEL 2: TF-IDF with RANDOM FOREST MODEL

1. Introduction

Baseline Model 2 is developed as an improvement over the initial baseline model by using a more advanced machine learning algorithm. While the first model used a simple linear classifier, this model introduces an **ensemble-based approach (Random Forest)** to better capture complex patterns in textual data.

The goal of this model is to enhance prediction performance and provide more robust results for sentiment or rating classification tasks.

2. Model Overview

This model combines:

- **TF-IDF (Term Frequency–Inverse Document Frequency)** for feature extraction
- **Random Forest algorithm** for prediction

By integrating these techniques, the model is able to process text data effectively and produce more stable predictions compared to simpler models.

3. Feature Engineering using TF-IDF

The textual data is transformed into numerical format using TF-IDF vectorization.

✓ Key characteristics:

- Assigns importance to words based on frequency and uniqueness
- Reduces the impact of commonly occurring words
- Converts each document into a high-dimensional feature vector

These numerical representations are then used as input for the Random Forest model.

4. Model Architecture – Random Forest

Random Forest is an **ensemble learning technique** that builds multiple decision trees and combines their outputs.

✓ Core concept:

- Multiple decision trees are trained on different subsets of data
- Each tree makes an independent prediction
- Final output is determined by:

    - Majority voting (classification)
    - Averaging (regression)

5. Working Mechanism

- Input text is converted into TF-IDF vectors
- Each decision tree learns different patterns from the data
- Random feature selection introduces diversity among trees
- Predictions from all trees are combined to produce the final output

This approach improves generalization and reduces the risk of overfitting.

6. Training Strategy

- The model is trained using labeled data
- Each tree is built using a random subset of features and samples
- This randomness helps the model learn diverse patterns

7. Model Performance and Observations

Compared to Baseline Model 1:

- Provides **better stability in predictions**
- Handles **non-linear relationships** more effectively
- Performs better in cases where text patterns are complex

However:

- Still struggles with understanding deep contextual meaning
- Performance depends heavily on feature quality

8. Advantages of the Model

- Handles complex and non-linear patterns
- Reduces overfitting through ensemble learning
- Works well with high-dimensional text features
- More robust than single-model approaches

Baseline Model 2 improves upon the initial model by using Random Forest, which enhances prediction capability and robustness. While it performs better than traditional linear models, it still lacks the ability to understand contextual relationships in text, highlighting the need for advanced deep learning approaches in subsequent models.

Output:

```
print("LINEAR REGRESSION RESULTS")
print("MAE:", mean_absolute_error(y_test, y_pred_lr))
print("MSE:", mean_squared_error(y_test, y_pred_lr))
print("R2 Score:", r2_score(y_test, y_pred_lr))
```

```
LINEAR REGRESSION RESULTS
MAE: 0.6776733393430431
MSE: 0.7479051889718326
R2 Score: -0.06951100290494372
```

```
print("\nRANDOM FOREST RESULTS")
print("MAE:", mean_absolute_error(y_test, y_pred_rf))
print("MSE:", mean_squared_error(y_test, y_pred_rf))
print("R2 Score:", r2_score(y_test, y_pred_rf))
```

```
...

RANDOM FOREST RESULTS
MAE: 0.629018745610098
MSE: 0.5986972382470612
R2 Score: 0.14385767988280418
```